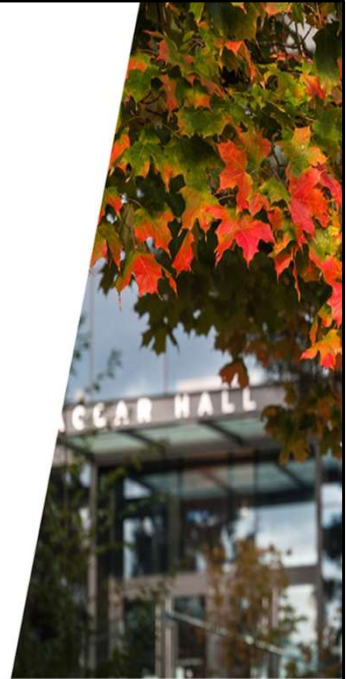




ADVANCED ANALYTICS AND FINAL PROJECT WORKSHOP

Data Analytics for Professional Accountants
(ACCTG 522)

Class 17 | MPAcc Class of 2026



Agenda

- **Welcome!**
- **Review: Final Project**
 - **Goal for today**
 - **Final Project Checklist**
 - **Final Project Appendices**
 - EDGAR Explorer and Extensions/Helpers
- **Final Project Workshop:**
 - Teams to work in assessment teams locally/remotely
 - Finalize any requests

Workshop classes

Goal facilitate completion of the final project.

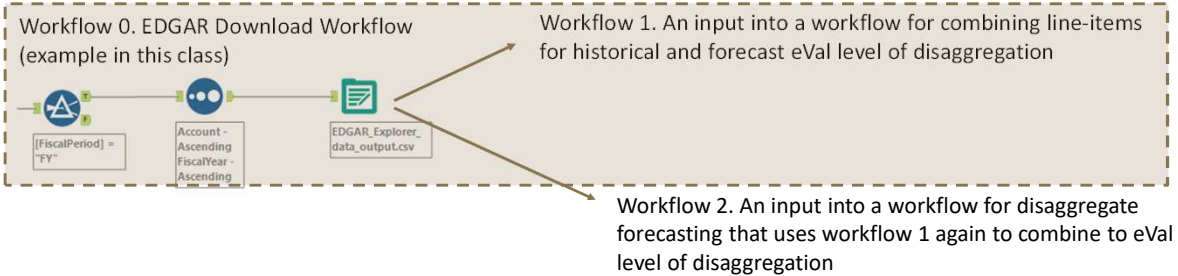
Class #	ACCTG 522	ACCTG521
18	11/24 Workshop	11/25 Blockchain (individual)
19	11/26 Thanksgiving break	11/27 Thanksgiving break
20	12/2 Workshop	12/2 Workshop *Audit Plan Deadline
21	12/3 Workshop *Final Project Due by 11:59PM	12/4 Final Presentations

Workshops do not require classroom presence, if teams want to meet remotely or in break out rooms, let me know.

Goals / benchmarking

Today: Goal is to complete ETL for historical data, e.g., in Alteryx:

Before Thanksgiving



Checklist

For each item you wish to enhance with data:

- Where can I get the data? Is it free? Do I have an API key? Are there restrictions on the data?
- Is the ETL easy or difficult to get into eVal? What kind of workflow do I need to use? Will the transpose be easy or difficult?

Checklist

Task	Access to Data?	ETL Comments
Replace eVal historical with EDGAR	EDGAR Explorer / SEC API	Data needs extraction using CFRAME and will need to be transposed to be pasted to eVal Financial Statements Tab
Replace eVal forecasts with Predictive tools based forecasts	Workflow to be developed in Alteryx	Leverages data above, need to better understand predictive tools.

Final Project Templates

Data Source Appendix

Goal: Have a summary Table documenting the sources of data used and where the data was used.

Appendix 1
Data Sources

Data Source	Used for
SEC-APL Company Facts	XBRL extracted data used as input to eVal Historical Financial Statement data
10-K: Item 7 MD&A	Additional risk analysis around changes in sentiment
10-K: Non-GAAP reconciliation	Assessing ratio sensitivity to non-GAAP ratios and alternative metrics

¶

Page Break ¶

Final Project Templates

Analytics Enhancement Appendix

Goal: Have a summary Table documenting the sources analytical methods used and where in the Financial Statement Analysis they were incorporated.

¶

Appendix 2¶

¶

Summary of Analytical Enhancements to FSA¶

Enhancement¶	Used to¶
Disaggregate forecasting of revenue¶	Separate individual revenue streams to better predict future Revenue¶
¶	¶
¶	¶
¶	¶

¶

..... Page Break¶

Final Project Templates

XBRL Mapping to eVal

Goal: Have a summary Table documenting the mapping of disaggregated tags to eVal line items.

¶

Appendix 3 (optional)¶
XBRL-Mapping to eVal¶

eVal-Line-Item¶	XBRL-Inputs¶
<i>Income-Statement¶</i>	
Sales (Net)¶	¶
Cost of Goods Sold¶	¶
Gross Profit¶	(Calculated)¶
R&D Expense¶	¶
SG&A Expense¶	¶
EBITDA¶	(Calculated)¶
Depreciation & Amortization¶	¶
EBIT¶	(Calculated)¶
Net Interest Expense¶	¶
Non-Operating Income (Loss)¶	¶
EBT¶	(Calculated)¶
Income Taxes¶	¶
Other Income (Loss)¶	¶
Net Income Before Ext. Items¶	¶
Ext. Items & Disc. Ops.¶	¶
Noncontrolling Interest in Earnings¶	¶
Preferred Dividends¶	¶
Net Income (available to common)¶	(Calculated)¶

¶

¶

Agenda

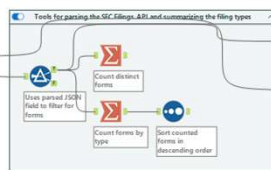
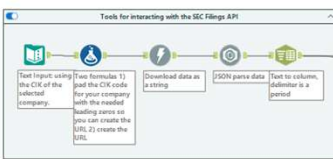
- **Welcome!**
- **Review: Final Project**
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Full text filing (text, tables, XBRL schema and more; *includes product/service axis*)

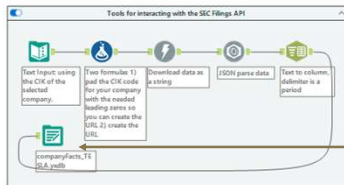
The steps here include using the input text tool to provide the URL for the download tool to connect to the API, the JSON data returned by the download tool is the parsed using the JSON tool which can then be deserialized further using the text to columns tool.

The steps here are to identify all of the components needed to answer the questions about the filings (the filter and summarize tools).

The tools in this step are used to identify the most recent 10-K filing components. First the components to construct the URL which can be used to download the most recent 10-K filing uses the filter tool, the sum and select records tools. Then using a join tool, and a formula tool, to construct the URL the download tool is used to interact with the SEC's website to download the most recent 10-K. A select tool and text to columns tool are then used to create a piv formatted 10-K which is saved using the output data tool.



Use this step first to download the SEC API data from company facts. It will provide output in the form of a join file with all of the XBRL data stored and indexed with some summary filings information (e.g., form filed 10-K/Q and cases).



JSON file of XBRL (largely numeric)

Usually at the aggregate level (does not include Product/Service Axis)

XBRL Tag

Tesla, Inc.
Consolidated Statements of Operations
(in millions, except per share data)

		Year Ended December 31,		
		2024	2023	2022
Revenues				
Automotive sales		72,480	\$ 78,509	\$ 67,210
Automotive regulatory credits		2,763	1,790	1,776
Automotive leasing		1,827	2,120	2,476
Total automotive revenues		77,070	82,419	71,462
Energy generation and storage		10,086	6,035	3,909
Services and other		10,534	8,319	6,091
Total revenues		97,690	96,773	81,462
Cost of revenues				
Automotive sales		61,870	65,121	49,599
Automotive leasing		1,003	1,268	1,509
Total automotive cost of revenues		62,873	66,389	51,108
Energy generation and storage		7,448	4,894	3,621
Services and other		9,921	7,830	3,880
Total cost of revenues		80,240	79,113	60,609
Gross profit		17,450	17,660	20,853
Operating expenses				
Research and development		4,540	3,969	3,073
Selling, general and administrative		5,150	4,800	3,946
Restructuring and other		684	—	176
Total operating expenses		10,374	8,769	7,197

Attributes

Revenue from Contracts with Customer, Excluding Assessed Tax

Tag	us-gaap:RevenueFromContractWithCustomerExcludingAssessedTax
Fact	72,480,000,000
Period	12 months ending 12/31/2024
Product and Service [Axis]	Automotive Sales (Member)
Measure	USD
Scale	Millions
Decimals	Millions
Balance	Credit
Sign	Positive
Type	monetaryItemType
Format	ixt:num-dot-decimal

XBRL Tag

Attributes	
Revenue from Contract with Customer, Excluding Assessed Tax	
Tag	us-gaap:RevenueFromContractWithCustomerExcludingAssessedTax
Fact	72,480,000,000
Period	12 months ending 12/31/2024
Product and Service (Axis)	Automotive Sales [Member]
Measure	USD
Scale	Millions
Decimals	Millions
Balance	Credit
Sign	Positive
Type	monetary/Item Type
Format	ixt:num-dot-decimal

XBRL Tag

Record	Account	Amount	Form	FiscalPeriod	FilingDate	FiscalPeriodEnd	FiscalYear	Frame
1	RevenueFromContractWithCustomerExcludingAs...	21461268000	10-K	FY	2019-02-19	2018-12-31	2018	CY2018
2	RevenueFromContractWithCustomerExcludingAs...	96773000000	10-K	FY	2024-01-29	2023-12-31	2023	[Null]
3	RevenueFromContractWithCustomerExcludingAs...	53823000000	10-K	FY	2024-01-29	2021-12-31	2023	CY2021
4	RevenueFromContractWithCustomerExcludingAs...	81462000000	10-K	FY	2024-01-29	2022-12-31	2023	[Null]
5	RevenueFromContractWithCustomerExcludingAs...	81462000000	10-K	FY	2025-01-30	2022-12-31	2024	CY2022
6	RevenueFromContractWithCustomerExcludingAs...	96773000000	10-K	FY	2025-01-30	2023-12-31	2024	CY2023
7	RevenueFromContractWithCustomerExcludingAs...	97690000000	10-K	FY	2025-01-30	2024-12-31	2024	CY2024

Same tag
no axis info

		2024	
Revenues			
Automotive sales		72,480	\$
Automotive regulatory credits		2,763	
Automotive leasing		1,827	
Total automotive revenues		77,070	
Energy generation and storage		10,086	
Services and other		10,534	
Total revenues		97,690	
Cost of revenues			
Automotive sales		61,870	
Automotive leasing		1,003	
Total automotive cost of revenues		62,873	
Energy generation and storage		7,446	
Services and other		9,921	
Total cost of revenues		80,240	
Gross profit		17,450	
Operating expenses			
Research and development		4,540	
Selling, general and administrative		5,150	

Attributes	
Revenue from Contract with Customer, Excluding Assessed Tax	
Tag	us-gaap:RevenueFromContractWithCustomerExcludingAssessedTax
Fact	97,690,000,000
Period	12 months ending 12/31/2024
Measure	USD
Scale	Millions
Decimals	Millions
Balance	Credit
Sign	Positive
Type	monetaryItem Type
Format	ixt:num-dot-decimal

eVal Considerations:

1. Five years of historical data is the default can ETL more but only paste in 5 years to avoid messing w/ eVal
2. Look for key missing data or other issues e.g., more important missing data

Company Name, Ticker and SIC	FESLA, INC.	FSLA	3711	Link to Form 10-Ks
Common Shares Outstanding	3,216,000	(in 000s at most recent fiscal year end)		Link to Industry Co
Estimated Price/Share=\$27.15				
	Actual	Actual	Actual	Actual
Fiscal Year End (YYYY-MM-DD)	2020/12/31	2021/12/31	2022/12/31	2023/12/31
	2024/12/31			
Income Statement				
Sales (Net)	31,536,000	53,823,000	81,462,000	86,773,000
Cost of Goods Sold	(24,906,000)	(40,217,000)	(60,609,000)	(79,413,000)
Gross Profit	6,630,000	13,606,000	20,853,000	17,660,000
R&D Expense	(1,491,000)	(2,593,000)	(3,075,000)	(3,969,000)
SG&A Expense	(2,746,000)	(4,075,000)	(3,629,000)	(4,800,000)
EBITDA	2,393,000	6,938,000	14,149,000	8,891,000
Depreciation & Amortization	(399,000)	(415,000)	(493,000)	0
EBIT	1,994,000	6,523,000	13,656,000	8,891,000
Net Interest Expense	(748,000)	(371,000)	(191,000)	(156,000)
Non-Operating Income (Loss)	(92,000)	191,000	254,000	1,238,000
EBT	1,154,000	6,343,000	13,719,000	9,973,000
Income Taxes	(292,000)	(699,000)	(1,132,000)	5,001,000
Other Income (Loss)	0	0	0	0
Net Income Before Ext. Items	862,000	5,644,000	12,587,000	14,974,000
Ext. Items & Disc. Ops.	0	0	0	0
Noncontrolling Interest in Earnings	(141,000)	(125,000)	(31,000)	23,000
Preferred Dividends	(31,000)	5,000	27,000	0
Net Income (available to common)	690,000	5,524,000	12,583,000	14,997,000

Less important

Asset classification and capital structure checks

Other assets is a plug,
e.g., missing investments

Implies Total Debt = 0

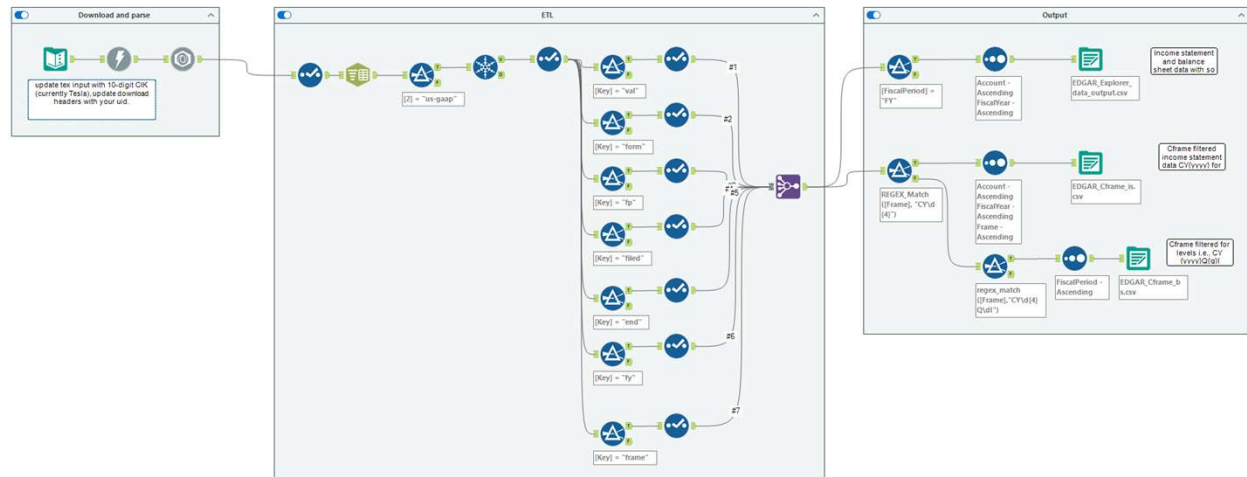
Other
candidates for
missing data
(less
important)

Balance Sheet					
Operating Cash and Market. Sec.	19,384,000	17,707,000	22,185,000	29,094,000	36,563,000
Receivables	1,895,000	1,924,000	2,952,000	3,508,000	4,418,000
Inventories	4,101,000	5,757,000	12,839,000	13,826,000	12,017,000
Other Current Assets	1,337,000	1,712,000	2,941,000	3,388,000	5,362,000
Total Current Assets	26,717,000	27,100,000	40,917,000	49,616,000	58,360,000
PP&E (Net)	12,747,000	18,684,000	23,548,000	29,725,000	35,836,000
Investments	0	17,707,000	22,185,000	0	0
Intangibles	520,000	457,000	409,000	431,000	394,000
Other Assets	12,164,000	(2,017,000)	(4,721,000)	26,846,000	27,480,000
Total Assets	52,148,000	62,131,000	82,338,000	106,618,000	122,070,000
Current Debt	1,758,000	1,088,000	1,016,000	0	0
Accounts Payable	6,051,000	10,025,000	15,255,000	14,431,000	12,474,000
Income Taxes Payable	777,000	1,122,000	1,235,000	0	0
Other Current Liabilities	5,662,000	7,470,000	9,203,000	14,317,000	16,347,000
Total Current Liabilities	14,248,000	19,705,000	26,709,000	28,748,000	28,821,000
Long-Term Debt	1,094,000	991,000	568,000	0	0
Other Liabilities	13,076,000	9,852,000	9,163,000	14,261,000	19,569,000
Deferred Taxes	0	0	0	0	0
Total Liabilities	28,418,000	30,548,000	36,440,000	43,009,000	48,390,000
Noncontrolling Interest	1,454,000	1,394,000	1,194,000	975,000	767,000
Preferred Stock	0	0	0	0	0
Paid in Common Capital (Net)	27,675,000	29,860,000	31,819,000	34,752,000	37,704,000
Retained Earnings	(5,399,000)	329,000	12,885,000	27,882,000	35,209,000
Total Common Equity	22,276,000	30,189,000	44,704,000	62,634,000	72,913,000
Total Liabilities and Equity	52,148,000	62,131,000	82,338,000	106,618,000	122,070,000

Analysis of Leverage
- Long-Term Capital Structure

Debt to Equity Ratio	0.128	0.069	0.035	0.000	0.000	0.000	0.000
FFO to Total Debt		1.150	6.765	25.343	#DIV/0!	#DIV/0!	#DIV/0!
CFO to Total Debt		2.799	5.530	26.941	#DIV/0!	#DIV/0!	#DIV/0!
Analysis of Leverage							
- Short-Term Liquidity							
Current Ratio	1.875	1.375	1.532	1.726	2.025	2.025	2.025
Quick Ratio	1.493	0.996	0.941	1.134	1.422	1.422	1.422
EBIT Interest Coverage	2.666	17.582	71.497	56.994	20.217	#DIV/0!	#DIV/0!
EBITDA Interest Coverage	3.199	18.701	74.079	56.994	20.217	#DIV/0!	#DIV/0!

EDGAR Explorer API data Helper



Other helpers/extractions

Hints on extraction from the full filing

Parse using known tags:
Document, Description,
Sequence etc.

FILER:

COMPANY DATA:
COMPANY CONFORMED NAME: Tesla, Inc.
CENTRAL INDEX KEY: 0001318605
STANDARD INDUSTRIAL CLASSIFICATION: MOTOR VEHICLES & PASSENGER CAR BODIES [3711]
ORGANIZATION NAME: 04 Manufacturing
IRS NUMBER: 912197729
STATE OF INCORPORATION: DE
FISCAL YEAR END: 1231

FILING VALUES:
FORM TYPE: 10-K
SEC ACT: 1934 Act
SEC FILE NUMBER: 001-34756
FILM NUMBER: 25570807

BUSINESS ADDRESS:
STREET 1: 3500 DEER CREEK RD
CITY: PALO ALTO
STATE: CA
ZIP: 94304
BUSINESS PHONE: 650-681-5000

MAIL ADDRESS:
STREET 1: 3500 DEER CREEK RD
CITY: PALO ALTO
STATE: CA
ZIP: 94304

FORMER COMPANY:
FORMER CONFORMED NAME: TESLA MOTORS INC
DATE OF NAME CHANGE: 20050222

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Thank you!

